

**10.** The crystallisation of sodium ethanoate from a super-saturated solution is used to release heat in reusable hand warmers.

- (a) A super-saturated solution of sodium ethanoate was made by dissolving 320 g of hydrated sodium ethanoate ( $\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}$ ) in  $60\text{ cm}^3$  of hot water. It was then allowed to cool to room temperature, which was measured as  $17^\circ\text{C}$ .

A thermometer was added to the solution, which caused the sodium ethanoate to start crystallising. The temperature of the process was recorded every 30 seconds for 3 minutes. The results are shown below:

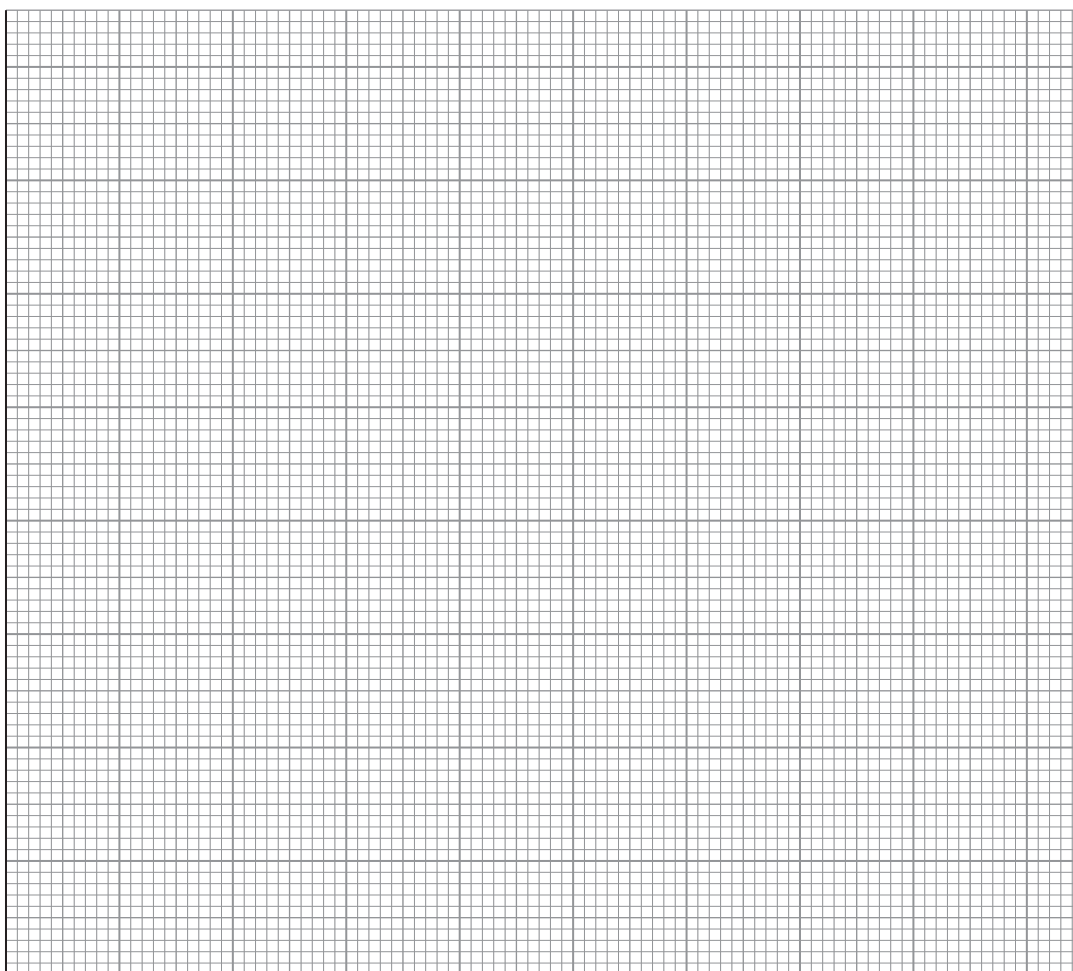
| Time/s | Temperature/ $^\circ\text{C}$ |
|--------|-------------------------------|
| 0      | 17                            |
| 30     | 27                            |
| 60     | 35                            |
| 90     | 41                            |
| 120    | 40                            |
| 150    | 39                            |
| 180    | 38                            |



- (i) Plot the results on the graph paper below.

[2]

Temperature / °C



Time / s

- (ii) Use your graph to calculate the maximum temperature change for this crystallisation.

[2]

maximum temperature change = ..... °C



- (iii) Use the **total mass** of the sodium ethanoate solution and the temperature change from the graph to calculate the enthalpy change of crystallisation per mole of sodium ethanoate. Assume the density of water is  $1.00 \text{ g cm}^{-3}$  and the specific heat capacity of sodium ethanoate solution is  $4.18 \text{ J K}^{-1} \text{ g}^{-1}$ .

$$M_r(\text{CH}_3\text{COONa} \cdot 3\text{H}_2\text{O}) = 136 \quad [4]$$

enthalpy change = .....  $\text{kJ mol}^{-1}$

- (iv) Suggest a reason why the experimental enthalpy change is often lower than the theoretical enthalpy change. [1]

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- (b) Sodium ethanoate can be made in a neutralisation reaction. Complete the following equation: [2]



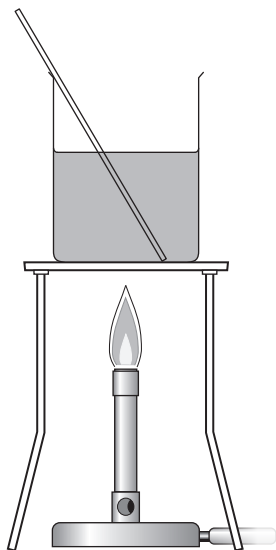
- (c) The carboxylic acid used to produce sodium ethanoate can be produced using an oxidation reaction.

- (i) Name the reagents and give the expected observations. [2]

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.....  
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- (ii) A student proposed that the apparatus below should be used to perform this oxidation reduction experiment.



The teacher said that this would not work and would be unsafe. Draw a labelled diagram of the apparatus that should be used in this experiment. [3]

END OF PAPER

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